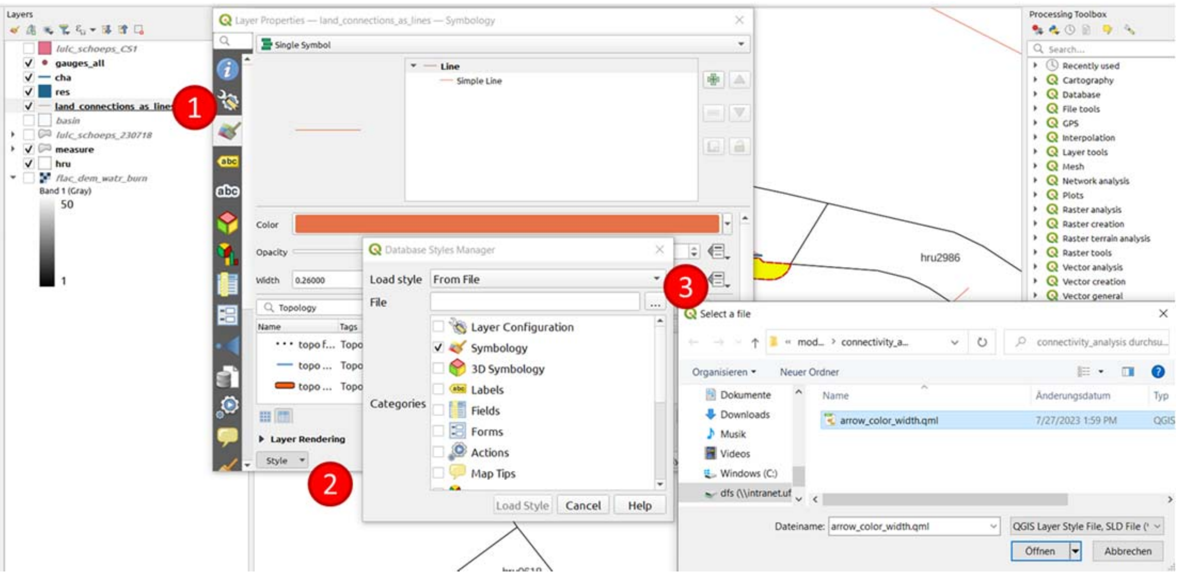
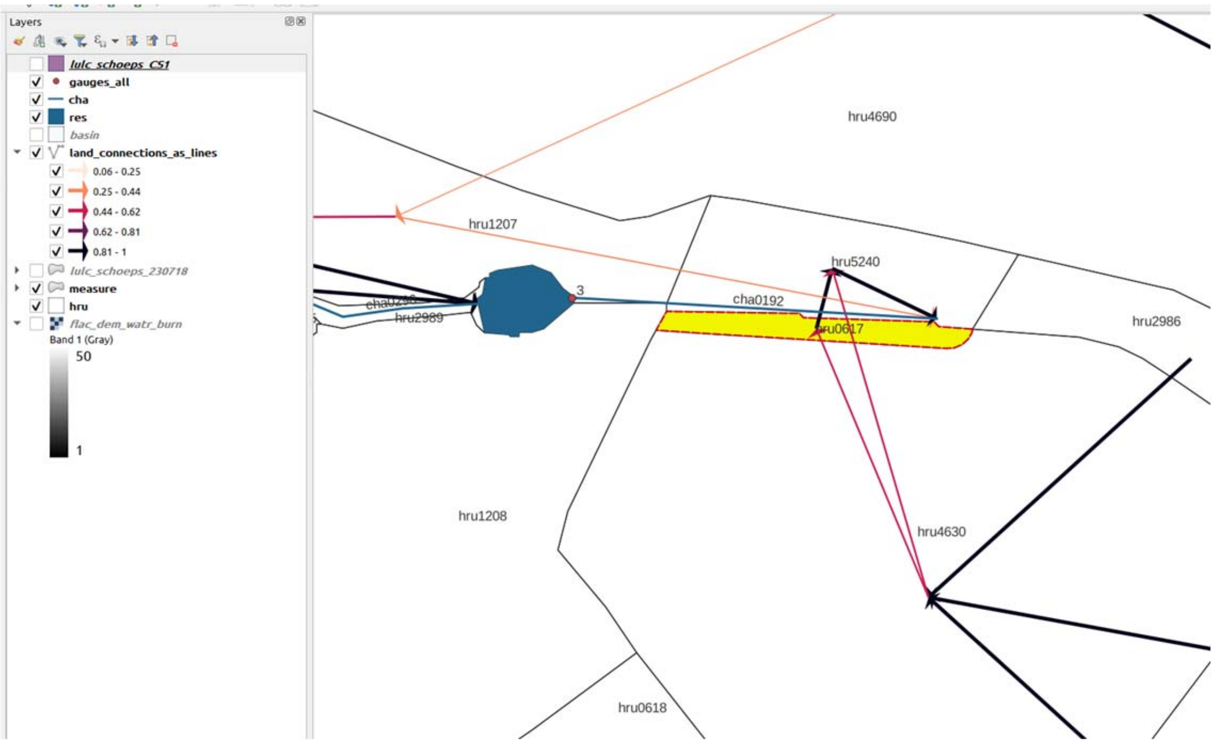
Check and correct connectivities – a small showcase (M. Strauch, 2023)

- connectivity fractions in your model setup were calculated based on your DEM (using the SWATbuildR)
- however, DEMs are not perfect and in some cases the calculated connectivities do not make any sense
- it is highly recommended to do a final check of connectivities before calibrating the model for streamflow und nutrients (at least check the connectivities for routing units that are supposed to represent NSWRMs in scenario simulations)
- all connectivities of land objects are defined in file rout_unit.con
- to visualize the connectivities run the script *create_connectivity_line_shape.R*
    - o input: paths to *data* folder created by SWATbuildR and the sqlite database
    - o output: land_connections_as_lines.shp
- open land_connections_as_lines.shp (from folder *data/vector*) and visualize connectivities in QGIS (see example below)



(1) Layer properties => Symbology
(2) Style => Load Style…
(3) Load style from file => Select file *arrow_color_width.qml*

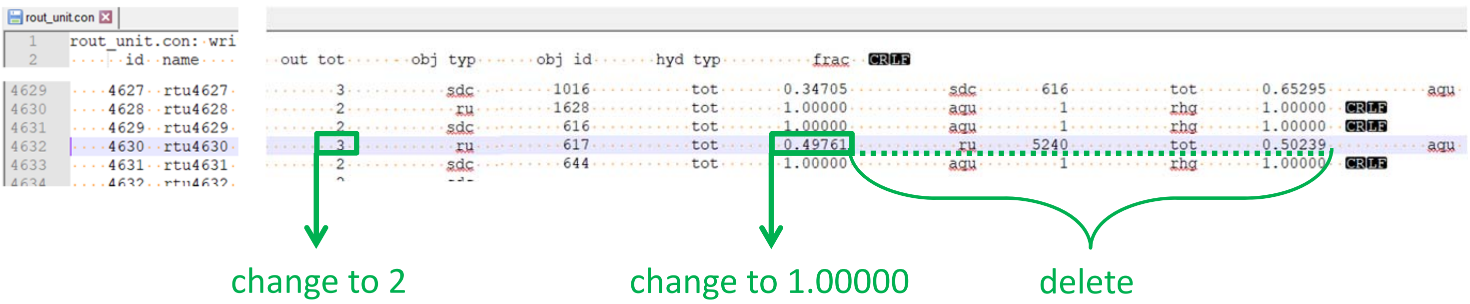
- Load also *hru.shp*, *cha.shp* and *res.shp* (from folder *data/vector*) and label them with their name
- Visualize routing units representing potential measures (e.g. using the landuse map where you defined those measures)
- For each potential measure, check if connectivities are plausible
- Below is an example of a potential riparian buffer (hru0617, yellow) with two connectivity issues



(1) Issue 1: hru4630 routes into two neighboring hrus (hru0617 and hru5240) with almost equal fractions (displayed by two red arrows). In fact, hru4630 and hru5240 have almost no shared border. The connectivity is an artefact of the DEM flow accumulation in the area where the edges of both units touch each other (at the beginning of the potential riparian buffer). It seems to be reasonable to remove their connection and instead route all water from hru4630 to the potential riparian buffer (hru0617). This correction will probably increase the performance of the riparian buffer (more water will be routed through this unit on the way to cha0192).

(2) Issue 2: the potential riparian buffer (hru0617) does not directly route any water into the closest channel (cha0192), it routes 100% (value 1) to hru5240. In our model setups, we should assume that riparian buffers have a direct connection to the channel network (if the buffer is upslope of the channel). In this case, it is recommended to replace the 100% connectivity to the land object (hru5240) with a 100% connectivity to the channel object (cha0192).

⇒ Solving Issue 1 (in file rout_unit.con, see example below): Delete the part specifying the connectivity to hru 5240 (do not accidentally delete the connectivity to the aquifer defined right after that part!), change the connectivity fraction with hru 617 to value 1 (=100%), and adapt total number of outgoing connections to value of 2



⇒ Solving issue 2 (in file rout_unit.con, see example below): Change object type from to sdc (channel) and change object id to 192 (name of the channel to route into)